Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *AFRIDOC_MT* | ***Context:*** *Amhara Region Agricultural & Microfinance Document Translation — Cooperative and Rural MFI End-Users*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Reference Amharic translations were produced by four named native-speaker translators [Q23, Q109] under a controlled workflow with QE filtering at threshold 0.65 [Q28, Q29] and language-coordinator manual inspection [Q27]. However, the user's preferred ground-truth authority is domain-trained federal bureau translators [elicitation Q3], with regional bureau staff or extension workers as acceptable surrogates — and no documentation of institutional affiliation or agricultural/legal domain expertise exists for the four Amharic translators [Q109; gap_3 web search NOT FOUND]. The translators were prepared for health/IT domains via a terminology workshop [Q26], not for proclamation-derived agricultural or financial vocabulary. Inter-annotator agreement for Amharic was Krippendorff's alpha = 0.46 [Q71, Q211], described by the authors as 'relatively low' [Q71], even within the health domain in which translators were specifically prepared. Reference translations also exhibit translationese due to English source provenance [Q103, Q104]. Human evaluation used three native-speaker annotators per language [Q194], primarily drawn from the translation team [Q194], 80 documents each with 30 shared [Q197]. No demographics on annotator regional origin (Amhara region vs. other Amharic-speaking areas) are documented. Given the user's HIGH priority weight on OC and the multiple convergent gaps (annotator domain mismatch, low IAA, translationese, no federal-bureau-translator anchor), score is 2.

ID	Evidence
Q3	“We conduct document-level translation benchmark experiments by evaluating the ability of neural machine translation (NMT) models and large language models (LLMs) to translate between English and these languages, at both the sentence and pseudo-document levels, the outputs being realigned to form complete documents for evaluation.” (p.1)
Q23	“We translated the extracted 10k English sentences to the 5 African languages through 4 expert translators per language.” (p.3)
Q26	“Due to the domain-specific nature of the task, before starting the translation process, we conducted a translation workshop, during which three translation experts shared their experiences in creating terminologies and they also shared existing resources with the translators including short translation guidelines (Appendix A.1).” (p.3)
Q27	“Quality control was conducted using automated quality estimation, followed by manual inspections by our language coordinators.” (p.3)
Q28	“We also used Quality Estimation (QE), specifically AfriCOMET (Wang et al., 2024a), Given a translated sentence in any African language and its corresponding source English sentence, AfriCOMET generates a score between 0 and 1, where 0 indicates poor quality and higher values signify better quality.” (p.3)
Q29	“The translators, in collaboration with the language coordinators, were tasked with reviewing instances that had quality estimation scores below 0.65.” (p.3)
Q71	“We obtained Krippendorff's alpha values of ≥ 0.40, which are relatively low due to the fine granularity of the evaluation scale.” (p.8)
Q103	“A potential limitation of our dataset is the influence of translationese (Koppel and Ordan, 2011). Since all source material translated originates in English, translated sentences in African languages may exhibit patterns such as unnatural syntax or overly literal phrasing.” (p.10)
Q104	“Although we have not conducted an analysis to quantify these effects, prior work suggests that they can affect MT model performance, generalization and evaluation including direct assessment (Freitag et al., 2019; Edunov et al., 2020).” (p.10)
Q109	“Also, we also appreciate the translators whose names we have listed below: 1. Amharic: Bereket Tilahun, Hana M. Tamiru, Biniam Asmlash, Lidetewold Kebede 2. Hausa: Junaid Garba, Umma Abubakar, Jibrin Adamu, Ruqayya Nasir Iro 3. Swahili: Mohamed Mwinyimkuu, Laanyu koone, Baraka Karuli Mgasa, Said Athumani Said 4. Yorùbá: Ifeoluwa Akinsola, Simeon Onaolapo, Ganiyat Dasola Afolabi, Oluwatosin Koya 5. Zulu: Busisiwe Pakade, Rooweither Mabuya, Tholakele Zungu, Zanele Thembani” (p.11)

Q194	"For the human evaluation, three native speakers of the African languages—primarily translators involved in the dataset creation—were recruited." (p.24)
Q197	"To assess consistency and inter-annotator agreement, 30 of the 80 documents were shared among all three annotators." (p.24)
Q211	"We obtained α scores of 0.46, 0.57, 0.40, and 0.81, and 0.54 for Amharic, Hausa, Swahili, Yorùbá, and Zulu respectively." (p.26)

Strengths

- Reference translations were produced by named native speakers under controlled workflow with QE [Q23, Q28]
- Pre-translation workshop shared terminology resources [Q26], demonstrating awareness of domain-terminology importance
- Translators worked with full document context [Q25], appropriate for document-level coherence requirements
- Translators were instructed not to use MT engines [Q120], reducing risk of MT-contaminated reference data
- Compensation was disclosed ($1,250 / 2,500 sentences for translators [Q31]; $55.15 for human evaluators [Q198]), enabling labor-conditions transparency

ID	Evidence
Q23	"We translated the extracted 10k English sentences to the 5 African languages through 4 expert translators per language." (p.3)
Q25	"The 10k sentences were distributed equally among the translators and the translations were done in-context (i.e. the translators translated on the sentence level but had access to the whole document)." (p.3)
Q26	"Due to the domain-specific nature of the task, before starting the translation process, we conducted a translation workshop, during which three translation experts shared their experiences in creating terminologies and they also shared existing resources with the translators including short translation guidelines (Appendix A.1)." (p.3)
Q28	"We also used Quality Estimation (QE), specifically AfriCOMET (Wang et al., 2024a), Given a translated sentence in any African language and its corresponding source English sentence, AfriCOMET generates a score between 0 and 1, where 0 indicates poor quality and higher values signify better quality." (p.3)
Q31	"Each translator was paid $1, 250 for 2, 500 sentences." (p.3)
Q120	"Use the language field to input the translations. It is essential not to rely on translation engines, as our quality assurance process can detect this. Depending on such tools may result in potential issues that you would need to address, leading to additional work on your part." (p.18)
Q198	"Each annotator was remunerated with $55.15" (p.24)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — ground-truth labels reflect health/IT translation judgments. The user-specified preferred authority (federal bureau translators with agricultural/legal Amharic training) is unrepresented [elicitation Q3; Q109 lists translator names without institutional affiliation; gap_3 web search confirmed no public information].

OC-2: Likely substantial disagreement on domain-specific term choices. Amhara regional bureau staff and federal Ministry of Agriculture translators would apply proclamation-derived standardized terminology [WEB-10, WEB-8]; AFRIDOC-MT translators applied health/IT register translations. Disagreement is structurally expected; the magnitude is not directly measured.

OC-3: Datasheet-level demographics absent. The paper names translators [Q109] and notes they were native speakers recruited via a coordinator [Q23, Q24] but provides no institutional affiliation, regional origin (Amhara-region vs. other), education, or domain training.

OC-4: Required for deployment validity. Re-annotation by Amhara Regional Bureau translation staff or federal Ministry of Agriculture translators on a deployment-domain test set is needed.

OC-5: Aggregation: human DA averages three annotators per language [Q69]. Krippendorff's alpha 0.46 for Amharic [Q71, Q211] indicates substantial inter-annotator divergence; averaging may erase legitimate domain-driven disagreements. Zulu evaluation could not be conducted at all due to recruitment difficulties [Q100], a precedent that flags fragility of the annotator-recruitment pipeline.

OC-6: Issues harming convergent validity: translators' health/IT focus likely fails to correlate with federal bureau translator judgment on agricultural/legal terminology [WEB-14 confirms no Amharic legal-administrative resources exist for cross-validation]. External validity: judgments do not generalize to deployment ground-truth authority. Empirical: low IAA [Q71], translationese [Q103], and no documented domain expertise [DATASET-MAJOR Concern 2].

ID	Evidence
Q3	"We conduct document-level translation benchmark experiments by evaluating the ability of neural machine translation (NMT) models and large language models (LLMs) to translate between English and these languages, at both the sentence and pseudo-document levels, the outputs being realigned to form complete documents for evaluation." (p.1)
Q23	"We translated the extracted 10k English sentences to the 5 African languages through 4 expert translators per language." (p.3)
Q24	"The translators were recruited through a language coordinator who is also a native speaker of the language." (p.3)
Q69	"In Table 10 we report average direct assessment (DA) scores (on a scale from 0 to 100) from three annotators per language for the health domain, when translating into four African languages." (p.7)
Q71	"We obtained Krippendorff's alpha values of ≥ 0.40, which are relatively low due to the fine granularity of the evaluation scale." (p.8)
Q100	"Therefore, we conducted a human evaluation for translations from English to African languages, comparing only three models due to cost constraints. However, we were unable to recruit annotators for Zulu." (p.10)
Q103	"A potential limitation of our dataset is the influence of translationese (Koppel and Ordan, 2011). Since all source material translated originates in English, translated sentences in African languages may exhibit patterns such as unnatural syntax or overly literal phrasing." (p.10)
Q109	"Also, we also appreciate the translators whose names we have listed below: 1. Amharic: Bereket Tilahun, Hana M. Tamiru, Biniam Asmlash, Lidetewold Kebede 2. Hausa: Junaid Garba, Umma Abubakar, Jibrin Adamu, Ruqayya Nasir Iro 3. Swahili: Mohamed Mwinyimkuu, Laanyu koone, Baraka Karuli Mgasa, Said Athumani Said 4. Yorùbá: Ifeoluwa Akinsola, Simeon Onaolapo, Ganiyat Dasola Afolabi, Oluwatosin Koya 5. Zulu: Busisiwe Pakade, Rooweither Mabuya, Tholakele Zungu, Zanele Thembani" (p.11)
Q211	"We obtained α scores of 0.46, 0.57, 0.40, and 0.81, and 0.54 for Amharic, Hausa, Swahili, Yorùbá, and Zulu respectively." (p.26)
WEB-3	EABC and primary cooperative unions are the institutional terminology anchors for deployment, distinct from health/IT translator preparation (https://addisstandard.com/sowing-under-shadows-of-violence-amhara-farmers-face-bleak-harvest-as-conflict-deepens-fertilizer-shortages-drives-up-input-prices/)
WEB-8	Cooperative Societies Proclamation 985/2016 defines the operative tier vocabulary (https://leap.unep.org/en/countries/et/national-legislation/cooperative-societies-proclamation-no-9852016)
WEB-10	Ministry of Justice provides authoritative Amharic terminology — federal bureau translators are the preferred authority unrepresented in the benchmark (https://justice.gov.et/en/law/cooperative-societies-proclamation-2/)
WEB-14	No public Amharic legal-administrative glossary resource exists (https://github.com/EthioNLP/Ethiopian-Language-Survey)

Information Gaps

- Institutional affiliation and domain expertise of the four named Amharic translators
- Regional origin of Amharic translators (Amhara region vs. Addis Ababa vs. diaspora)
- Whether any of the human evaluators had agricultural or legal Amharic exposure

Requires Expert Verification

- Federal bureau translator judgment on AFRIDOC-MT reference Amharic register-appropriateness for agricultural/legal documents
- Magnitude of expected disagreement between AFRIDOC-MT translators and federal bureau translators on a deployment-domain test set

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Remediation

Gap 1: Reference Amharic translations were produced by translators with no documented agricultural or legal Amharic expertise; user-specified preferred authority (federal bureau translators) is absent; Amharic IAA is 0.46 even within the prepared health domain.

Recommendation: For deployment validation, recruit at least two federal Ministry of Agriculture or Amhara Regional Bureau translators to re-annotate a deployment-domain test set. Document annotator institutional affiliation, agricultural/legal Amharic experience, and regional origin. Report Krippendorff's alpha disaggregated by term-category to surface terminology-specific disagreement rather than averaging it away.